Reln l algebra - aggregation

- combine rows into fewer ones

$\gamma_{max(servings)}(A)$

10

$rid\ \gamma_{max(servings)}(A)$

find max for each distinct rid

rid	max(servings)
1	2
2	10
3	5
4	3
5	2.5

How SQL works

select ____ ← attributes

from ____ ← relations

where ____ ← conditions

```
select name
from recipes
where id = 2;
```

→ PerfectPopcorn

```
select R.name
from recipes R, assignments A
where assignments.iid = 3 and
      R.rid = A.rid
```

$$\Pi_{R.name}\left(\sigma_{A.iid=3 \wedge R.rid=A.rid}(R \times A)\right)$$

$$\Pi_{R.name}\left(\sigma_{A.iid=3}\left(R \bowtie_{R.rid=A.rid} A\right)\right)$$

select ——— Π

from ——, ——, ——— $\times$

where ——— and ——— and ——— σ

We did a join via specify tables in "from" and using "where" to subset rows.

Newer approach:

```
select R.name
from recipes R
join assignments A
  on R.rid = A.rid
where A.iid = 3
```

```
SELECT DISTINCT I.iid, I.name
FROM ingredients I
JOIN assignments A
ON I.iid=A.iid
JOIN recipes R
ON A.rid=R.rid
WHERE R.name = 'Spiced Pumpkin Soup'
AND R.name = 'Sugar Cookie Icing';
```

Is name both Sp Pum Soup and Sugar Cookies Icing

# of tuples is 0.

After joins

R.rid	name	...	aid	A.rid	iid	Sugs	...
1	Sp Pum Soup		1001	1	0	2	...

OR is probably what was desied here.

INSERT INTO recipes VALUES
(10, 'Awesome Pizza', NULL, 2000);
INSERT INTO recipes VALUES
11 (~~10~~, 'Easy Ice Cream', NULL, 1922);
After doing so, how many tuples would result from the following query?
SELECT * FROM recipes WHERE rating = NULL;

Answer: 0 tuples. WTF?

NULL in SQL is messy.

NULL is supposed to represent missing data.

- not an actual value.
- nothing is ever supposed to be equal to null.

"is null"

all attrs

```
select * from recipes
  where rating is null;
```

↳ 2 tuples

an equality test with null returns null

```
SELECT A.aid
FROM assignments A
WHERE A.servings > ANY (SELECT A2.servings
                        FROM assignments A2
                        WHERE A2.iid=1);
```

greater than any, meaning any one,

so get all rows w/ servings > 0.5

↳ results in 7 rows

servings
0.5
2.5

2 rows

Aggregate queries

select max(R.year)
from recipes R

```
select A.iid, max(A.servings)
from assignments A
group by A.iid;
```

one row for each iid, and then combine all rows w/ that iid using the aggregation function

```
select A.iid, count(*)
from assignments A
group by A.iid
having count(*) >=2;
```

WTF is the diff between WHERE and HAVING?

WHERE → works on data from the tables

HAVING → works on aggregates

SQL does in this order:

1. Runs select/from/join/where
2. Aggregates
3. Runs HAVING clauses

→ Or, at least, this is how you understand what the results must be

```
SELECT I.name
  FROM ingredients I
 WHERE EXISTS
       (SELECT A.iid
        FROM assignments A
        WHERE A.rid=4
          AND I.iid=A.iid);
```

subquery

→ has a condition depending on a row in the outer query

Correlated subquery